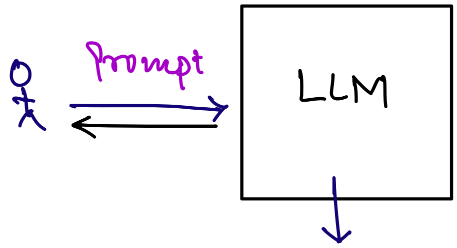

Pre-trained model

[It is trained on huge amount of internet data]

⇒ Hallucination.

- Not desirable
- Try to reduce hallucination as much as possible.

⇒ LLMs are very powerful but they have limited knowledge.

Limitations of LLM :

→ Static knowledge.

→ Doesn't have access to private policy documents.

→ Hallucination.

Grounded AI Systems.

Model's response should be based on external, verifiable, reliable information & not only on internal memory

RAG.

↳ Retrieval Augmented Generation.

It is an approach where the system :

→ takes the user query

→ retrieves the information from external knowledge sources.

→ Provide this Information to the LLM as context.

→ LLM generates the answer using the retrieved context.

RAG combines retrieval of relevant information from external sources & generation of responses in natural language.

So, instead of answering only from memory, the system answers using external sources as well.

RAG helps solve the main limitations of plain LLM.

"Grounding" in RAG.

- Reduces Hallucinations.
- More reliable, accurate & trustworthy answers.
- Makes it easier to trace the source of the information.

RAG flow.

⇒ We have already stored data in the form of embeddings in Vector DB.

Query → Convert the query into Embedding → Similarity Search.

↓

Provide the Context to LLM

↓

Generate & return the response.

	LLM-Only	RAG Based
Knowledge Source	Only pre-trained memory	pre-trained memory + external source
Access to latest info	Usually No	Yes
Access to Private info	No	Yes
Hallucination	High	Very low.

RAG + Vector DB.

⇒ RAG based applications used Vector DB to store & retrieve information.

- Exact Match
- Similarity Search.

⇒ Modern AI applications uses combination of Exact match & Similarity search for better responses.

#

Documents

↓

Split into Chunks.

↓

Create Embeddings

↓

Store in Vector DB

↓

User Query

↓

Create Embedding

↓

Similarity Search

↓

Retrieve Chunks.

↓

Pass to LLM

↓

Generate Answer.

RAG in Agentic AI

→ Agentic flow

Goal

↓

Agent perceives.

↓

Information ⇒ LLM + External Source.

↓

Action.

the informal